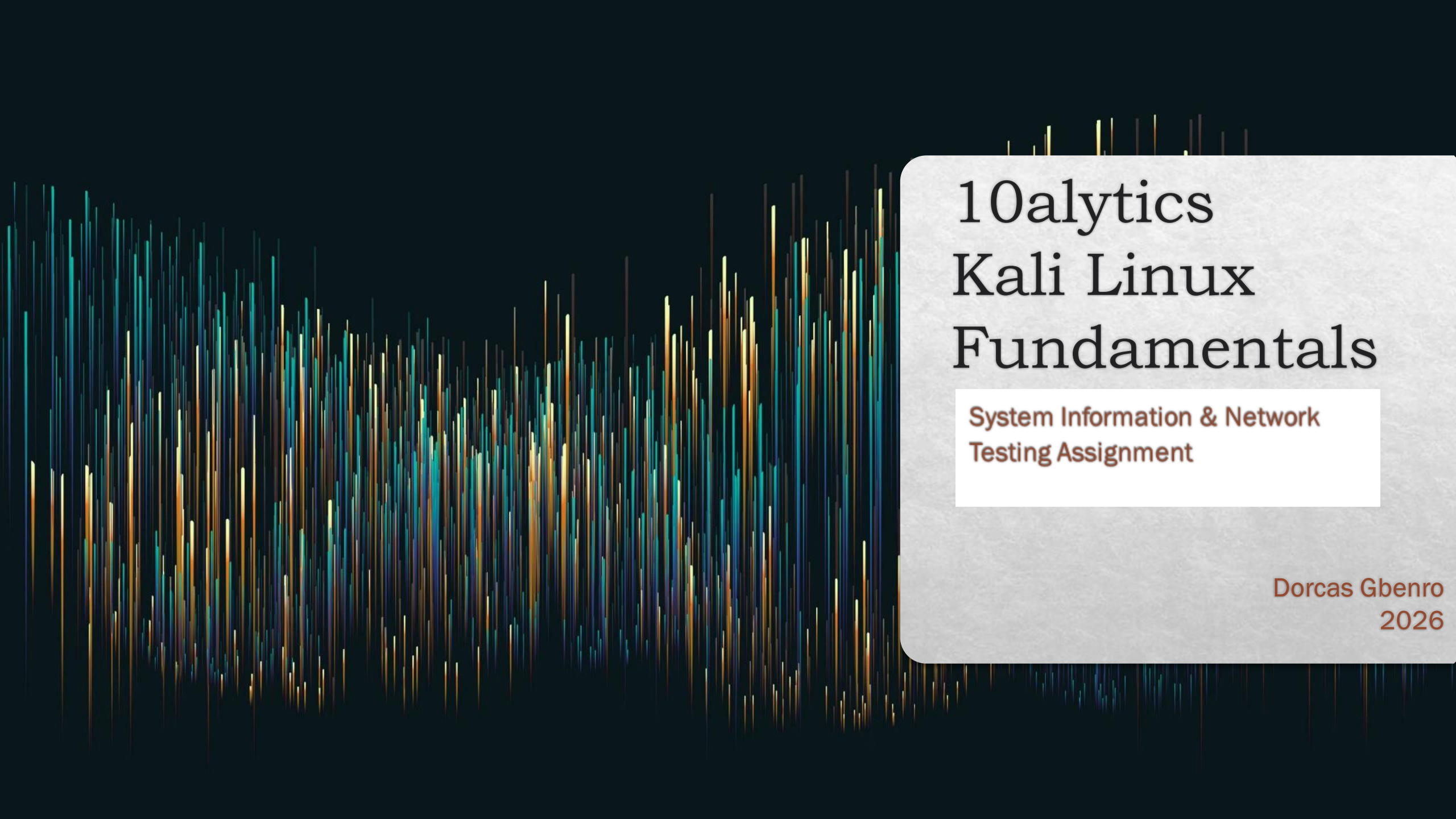


10alytics Kali Linux Fundamentals

System Information & Network Testing Assignment

Dorcas Gbenro
2026

Part 1 – Creating Directories

Requirements

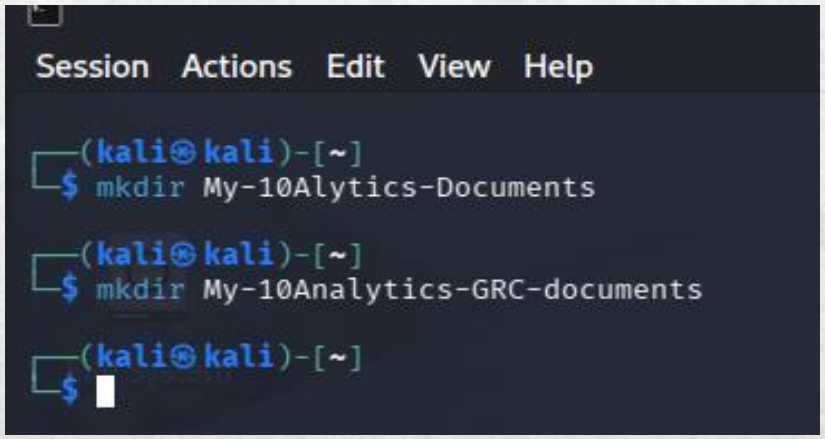
Cybersecurity Track Requirements

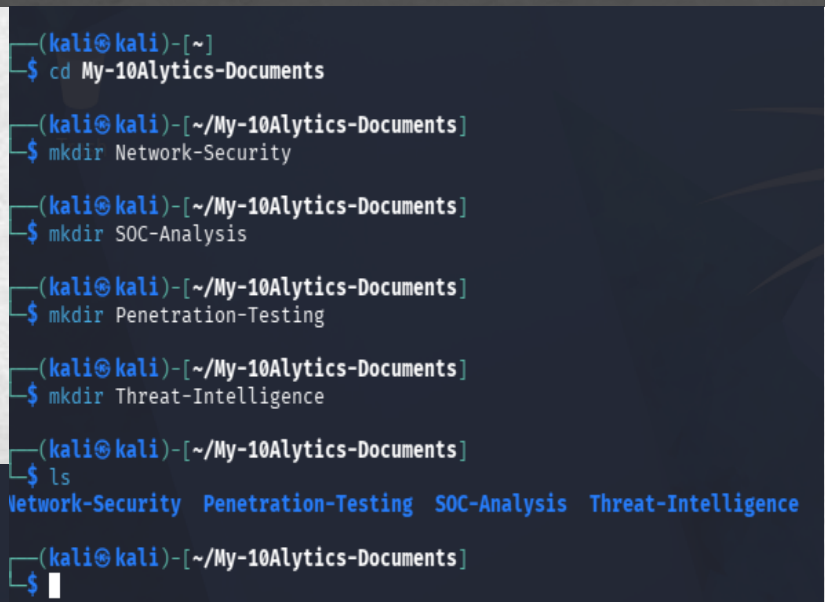
1. Create root folder: "my 10Aytics documents"
2. Create subfolders inside root:
   - Network Security
   - SOC Analysis
   - Penetration Testing
   - Threat Intelligence
3. Use **nano** and **mousepad** to create a **.txt** file within one of the directory.

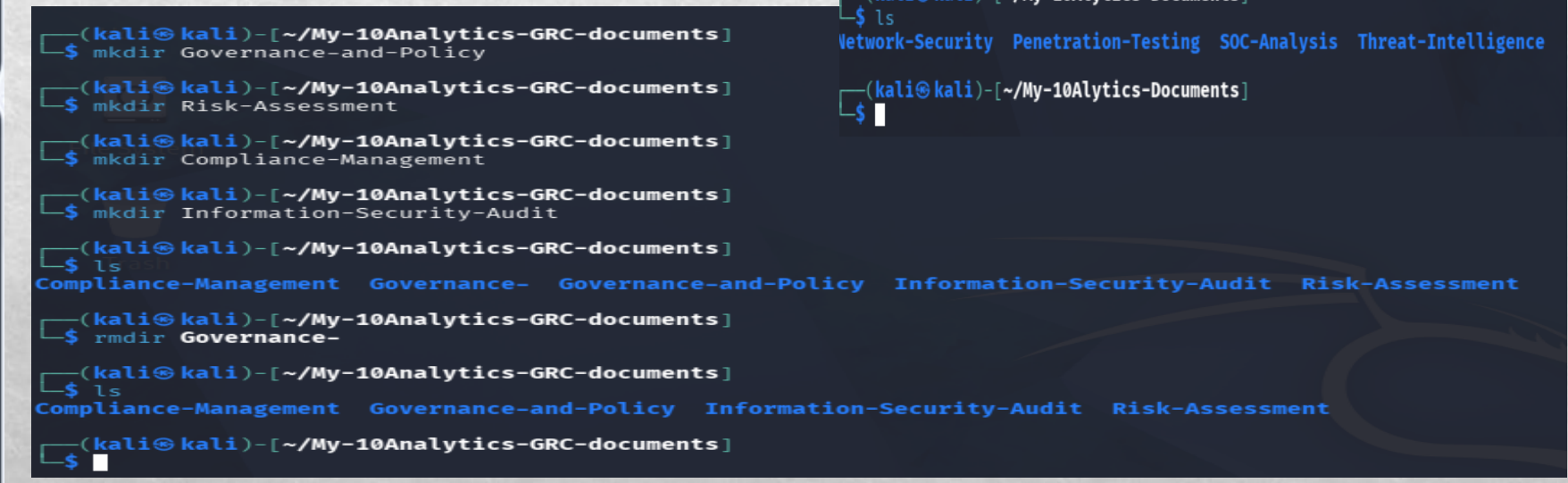
GRC Track Requirements

1. Create root folder: "my 10Alytics GRC documents"
2. Create subfolders inside root:
   - Governance & Policy
   - Risk Assessment
   - Compliance Management
   - Information Security Audit
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Screenshots









Part 1 – Creating Directories

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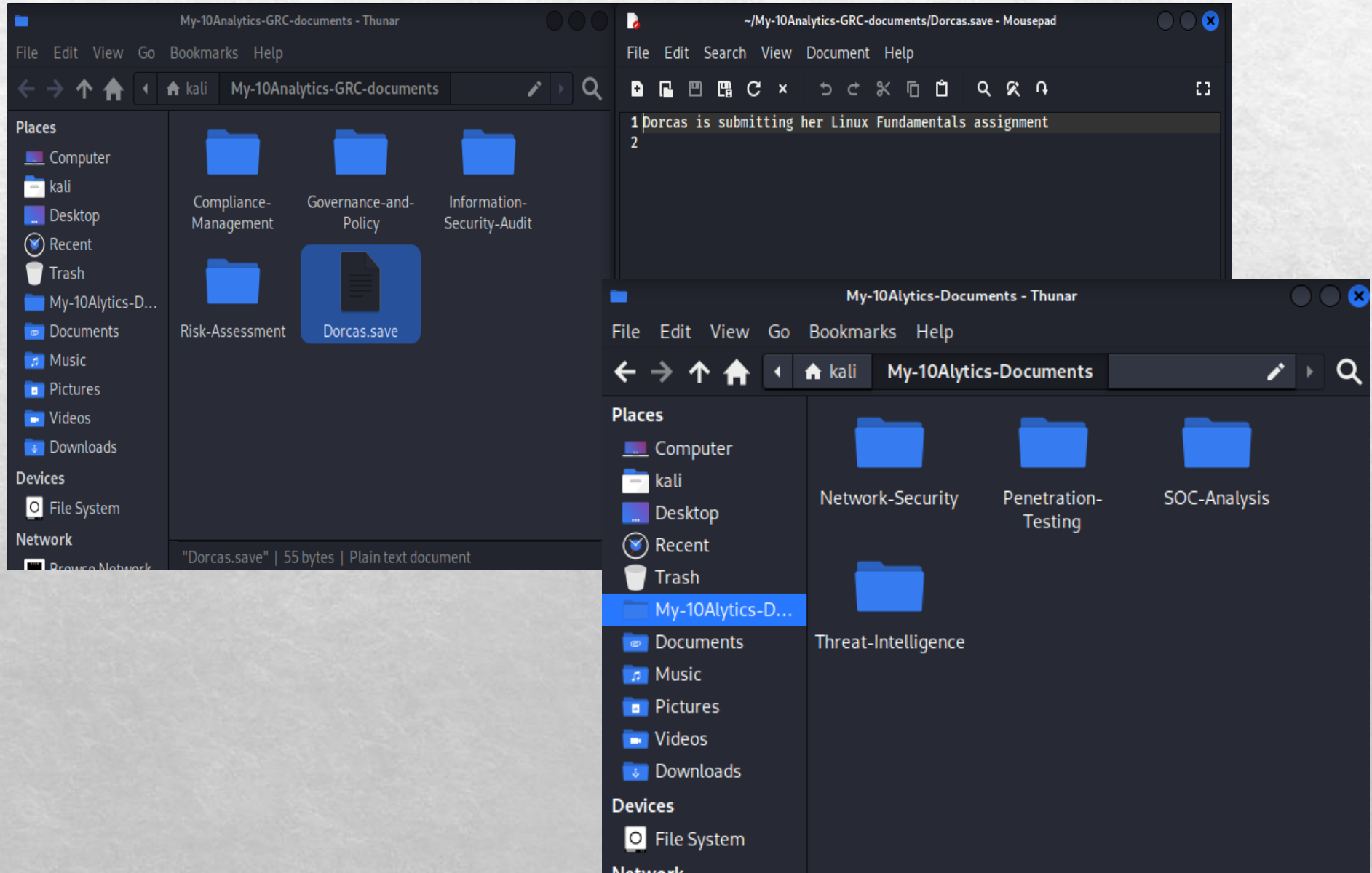
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Full Folder view



Part 2 – System Information

Requirements

Step 1 – Who am I and where am I?

Run the following commands one at a time in your terminal:

`whoami` # Displays the currently logged-in username

`hostname` # Displays the name of your machine

`pwd` # Shows which directory you are currently in

Step 2 – Operating System & Hardware Info

Use the commands below to learn more about your system. Try them out and note what each one tells you:

`uname -a` # Full system/kernel information

`lscpu` # CPU details (type, cores, speed)

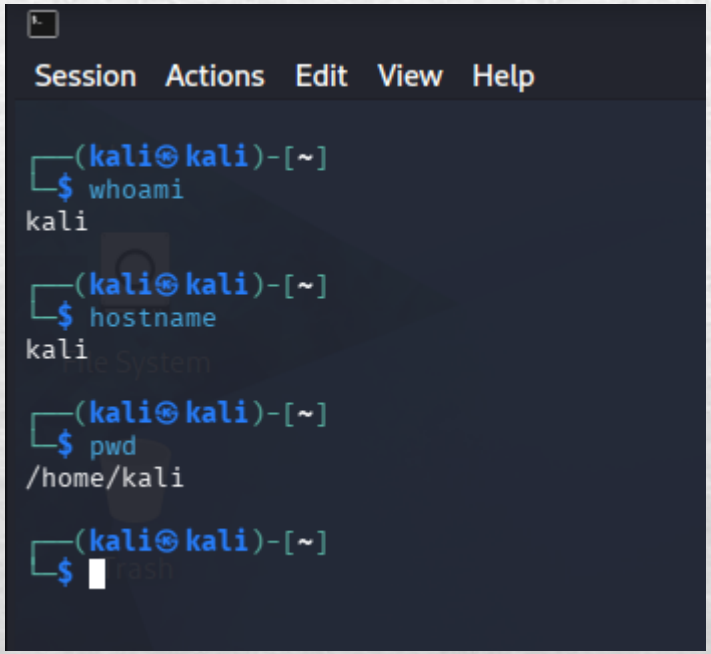
`free -h` # RAM usage in human-readable format

`df -h` # Disk space usage per partition

`uptime` # How long the system has been running

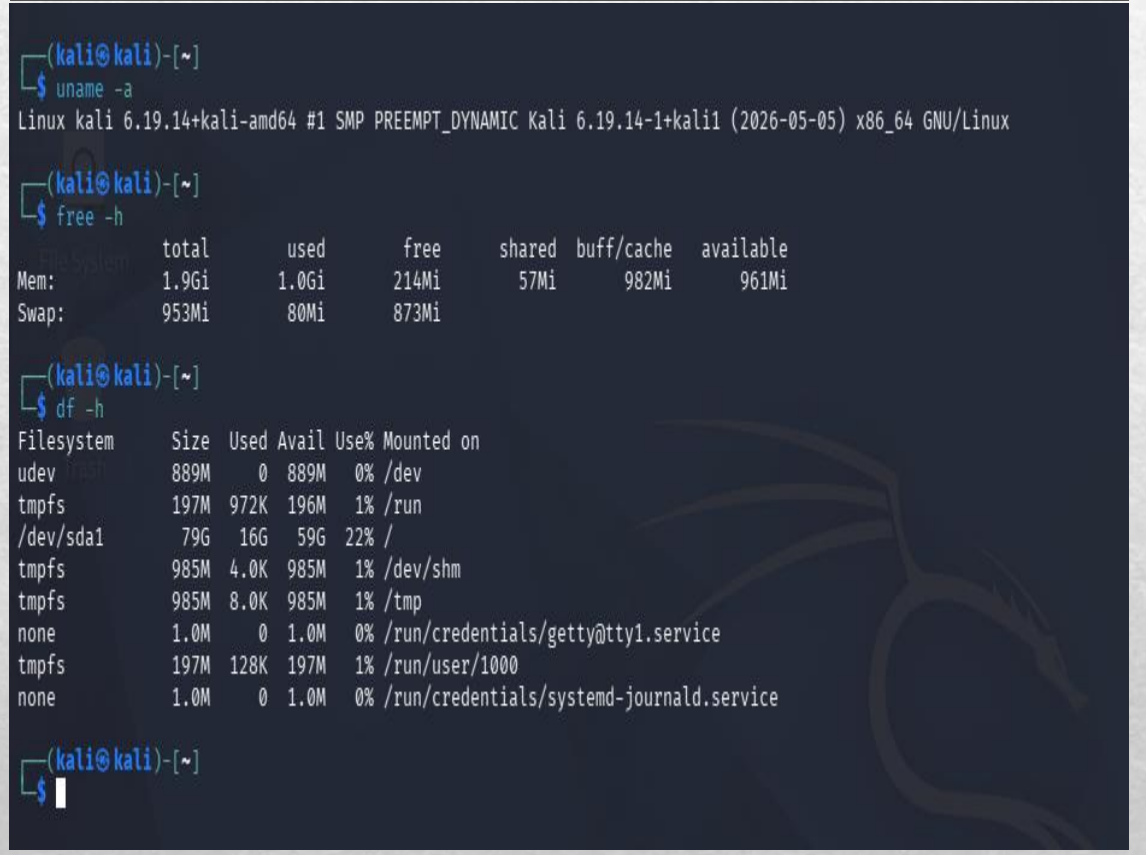
Screenshots

Step 1



- UserName: kali
- HostName: kali
- Current Directory: /home/kali

Step 2



- Total RAM: 1.9Gi
- Disk Space: 889M
- Linux Kernel Version: Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux

Part 3 – Basic Network Connectivity Test

Requirements

Step 1 – Ping a known IP Address

The IP address 8.8.8.8 belongs to Google's public DNS server. It is always online and is commonly used to test basic internet connectivity:

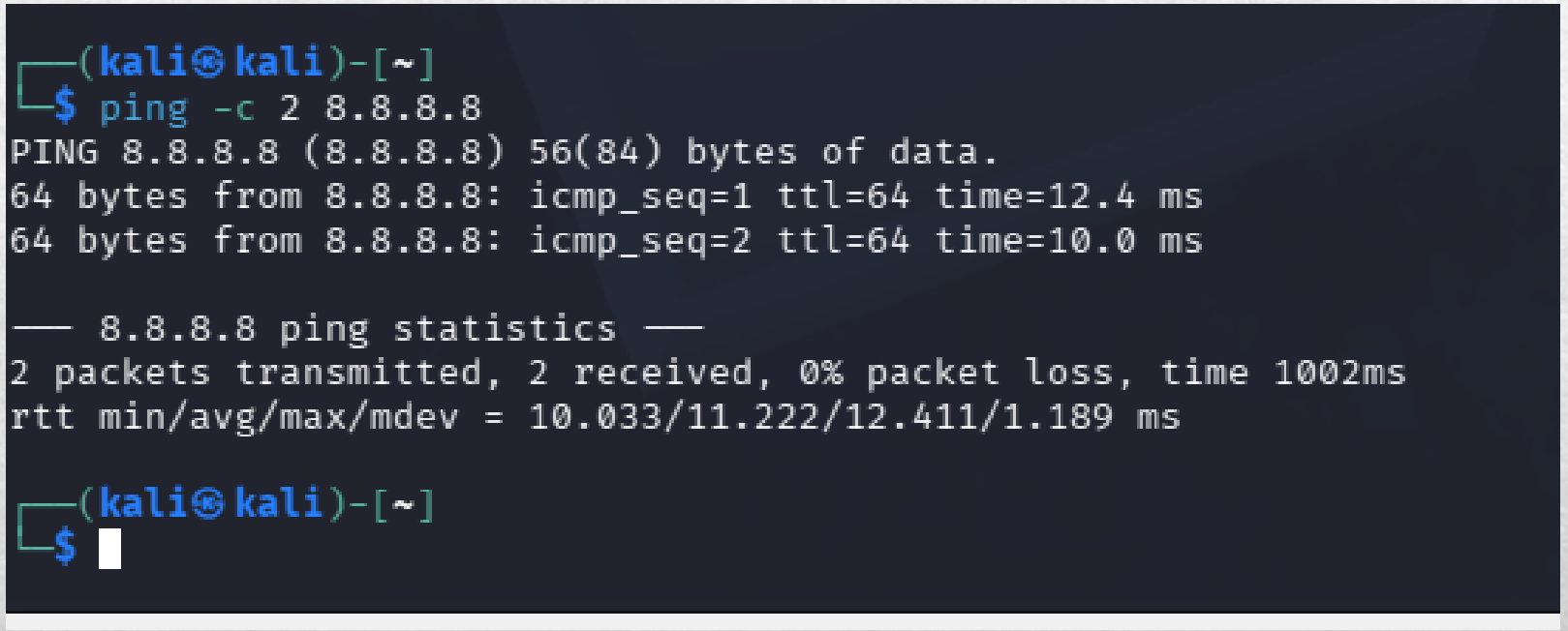
```
$ ping -c 2 8.8.8.8

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=12.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=11.8 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss
```

Screenshots

Step 1



- Average roundtrip time: 11.222
- Packet loss: 0%

Part 4 – Basic Network Connectivity Test

Requirements

Step 2 – Ping a Domain Name

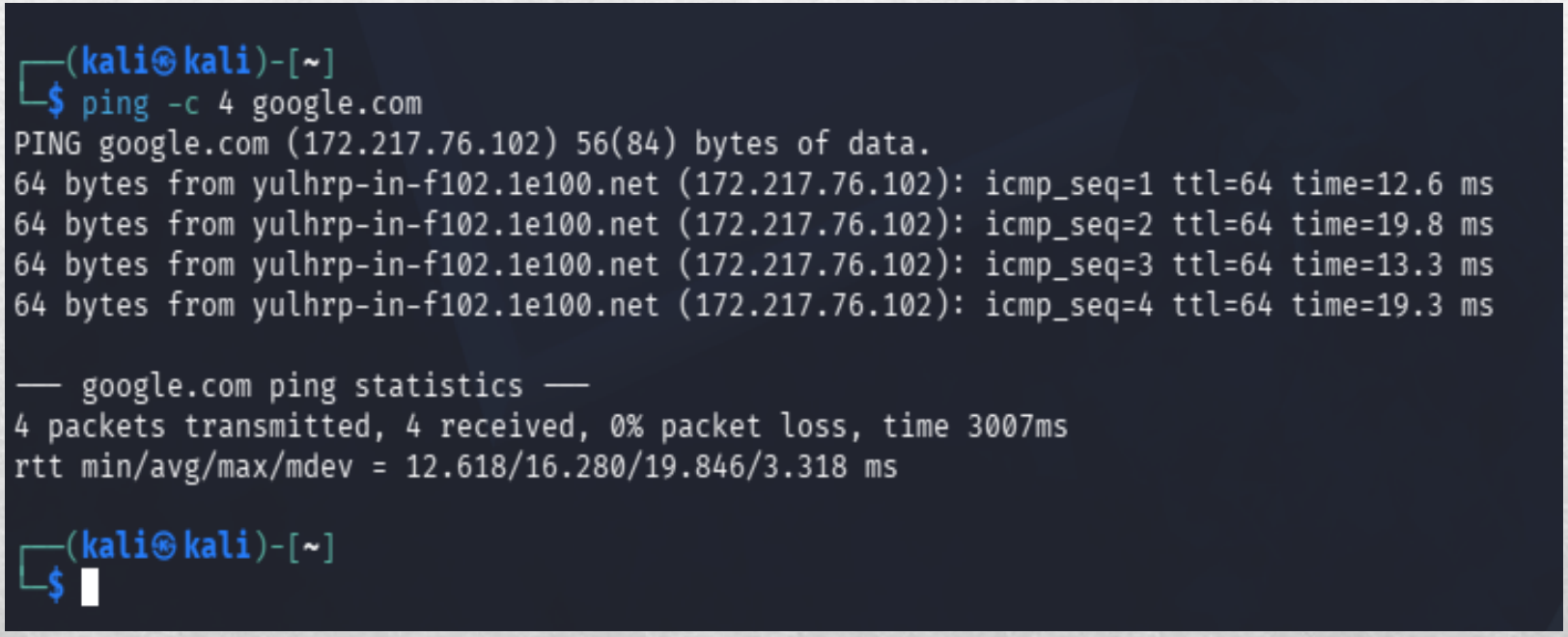
Now ping a website by name. This tests both internet connectivity *and* DNS resolution (translating a name like google.com into an IP address):

```
$ ping -c 4 google.com
```

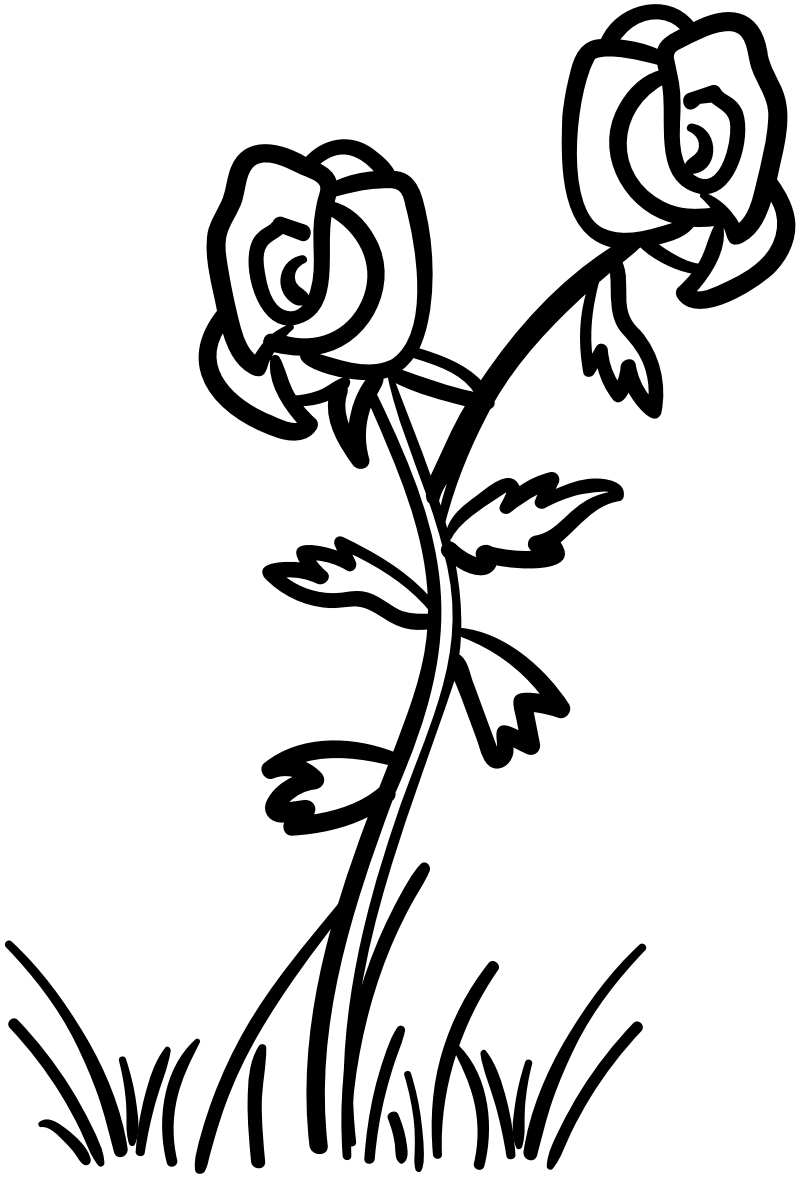
Notice: the terminal resolves the name to an IP address automatically

Screenshots

Step 2



- IP Address of google: 172.217.76.102
- The round trip time is different from pinging 8.8.8.8 directly
- The difference is because it took an additional time to generate the IP address from the domain name input.
- C-4 (Count 4) means the fourth count of packets in a ping command.



Thank you!